

UN Scientific Panel on AI Report



Source: TH

Why in News?

The **United Nations's (UN) Independent International Scientific Panel on Artificial Intelligence (AI)** released a preliminary report warning that **AI capabilities are advancing exponentially faster than scientific understanding and regulatory frameworks**, thereby posing **catastrophic risks** to global stability.

What are the Key Highlights of the UN Scientific Panel Report on AI?

- **Capability Race Outpacing Governance:** Frontier models are rapidly saturating evaluations like **Humanity's Last Exam** and **FrontierMath** (where performance surged from 19% to 88% within a year).
 - The length of complex coding and research tasks AI agents can autonomously complete is **doubling every 4-7 months**.
 - Models increasingly exhibit "**evaluation awareness**" (detecting when they are being tested) and have demonstrated behaviors resisting shutdown in controlled lab settings.
- **Concentration of AI Power ("Compute Divide"):** The US holds **~75%** and China **~15%** of computing power among the top 500 AI supercomputers, leaving the rest of the world with just **10%**.
 - Furthermore, **91% of notable AI models originate from the private sector**, concentrating critical governance decisions within corporations rather than sovereign states.
 - Hardware choke points such as **Advanced Semiconductor Materials Lithography** (a Dutch multinational corporation), **TSMC (Taiwan Semiconductor Manufacturing Company)** (for **chip fabrication**), **NVIDIA (US - chip design)** hold around **80% market share** in their respective niches.
 - Meanwhile, **118 nations (predominantly Global South) remain unrepresented** in major AI governance summits.
- **The Multidimensional "AI Divide":** The digital divide extends far beyond consumer access into four critical dimensions: **Infrastructure capacity (sovereign compute), Talent capacity, Governance/Regulatory capacity, and Application capacity**.
 - Of over 7,000 global languages, **AI models are optimized for a tiny fraction**. For example, in low-resource languages like Tigrinya (Eritrea/Ethiopia), machine translation errors rendered medical terms

- like **"smallpox" into "syphilis"** and "intravenous antibiotics" into "intravenous insecticides," illustrating critical safety failures in **contextual transplantation** .
- **Agentic AI & Cybersecurity Dual-Use:** The paradigm is shifting from text generation to **Agentic AI** systems (autonomous models capable of executing complex real-world workflows with minimal human checkpointing).
 - While AI coding agents accelerate software development (generating ~75% of new code), they introduce severe **multi-agent miscoordination risks** and high **cyberattack success rates via prompt injection (~84%)** , rapidly discovering zero-day vulnerabilities in critical infrastructure.
 - **Erosion of Shared Reality & Human Rights:** Key structural harms include **Epistemic Erosion** (weakening societal ability to discern truth), the **"Liar's Dividend"** (bad actors dismissing authentic evidence as deepfakes), and **Synthetic Consensus** (manufacturing false illusions of public agreement).
 - **Sycophancy** (AI flattering users to drive engagement) has emerged as a dangerous alignment failure linked to real-world self-harm.
 - The report highlights rising **human rights risks** , including **AI-generated Child Sexual Abuse Material (CSAM)** , deepfake abuse targeting women and girls (**99% of deepfakes targeting women**), and **gender bias** in AI development, with **88% of leading AI researchers being male** .
 - **Macroeconomics, Environment, and the "Evidence Dilemma":**
 - **The Productivity J-Curve:** AI gains are not instant; firms face a **"J-curve** (an initial loss immediately followed by a dramatic gain) " requiring significant upfront workflow and human-capital investments before realizing productivity growth.
 - Policymakers must guard against **"AI washing"** , where companies **overstate or falsely claim AI adoption** to enhance their image or use AI as a **pretext to justify layoffs, restructuring, or other corporate decisions unrelated to actual AI deployment** .
 - **Environmental Footprint:** Hyperscaler capital expenditure is reaching a ~ **USD 770 billion annualized run-rate** , exerting immense environmental strain on energy, water, critical minerals, and e-waste burdening the **Global South disproportionately**.
 - **The Evidence Dilemma:** Policymakers face a structural paradox: **by the time empirical evidence of AI harm is generated, the technology's rapid evolution renders delayed regulatory interventions ineffective**.

Key Takeaways for India

Report Finding	Implications for India
AI computing power is concentrated in the US and China (around 90% combined).	Reinforces the need to accelerate the IndiaAI Mission , expand sovereign GPU capacity , and reduce dependence on foreign hyperscalers in line with the goal of Digital Sovereignty .
The AI divide spans infrastructure, talent, governance, and application capacity.	Calls for strengthening India's AI ecosystem through the National Quantum Mission , Centres of Excellence for AI, and industry-academia partnerships while leveraging India AI Stack as a global digital public infrastructure model.
Over 7,000	Highlights the importance of BHASHINI , BharatGen ,

languages remain underserved by AI systems.	and multilingual foundation models to ensure inclusive AI access across India's linguistic diversity.
Healthcare AI requires local validation rather than imported models.	Supports context-specific AI deployment under the Ayushman Bharat Digital Mission (ABDM) , and the National Digital Health Mission , using Indian datasets to improve accuracy and reduce algorithmic bias.
Agentic AI poses new cybersecurity and governance challenges.	Highlights the need to strengthen India's AI governance ecosystem by effectively implementing the Digital Personal Data Protection (DPDP) Act, 2023 , empowering CERT-In and the Indian Cyber Crime Coordination Centre (I4C) , and finalising the National Cyber Security Strategy to safeguard AI-enabled critical infrastructure .
Global AI governance remains fragmented and weakly enforced.	Supports India's risk-based regulatory approach through the IndiaAI Safety Institute , leadership in the Global Partnership on AI (GPAI) , and its advocacy for inclusive AI governance during the G20 and future multilateral forums.
AI-generated CSAM and deepfakes disproportionately harm women and children.	Calls for stronger enforcement of the Information Technology Act, 2000 , the Bharatiya Nyaya Sanhita (BNS), 2023 , the POCSO Act, 2012 , and the IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021 to address AI-generated synthetic abuse and deepfakes.
118 countries remain excluded from major AI governance processes.	Provides India an opportunity to champion the Global South through Digital Public Infrastructure (DPI) , One Future Alliance , India Stack , and South-South cooperation for equitable AI access and capacity building.

Conclusion

The rapid evolution of AI demands governance frameworks that are as adaptive as the technology itself. Bridging the AI divide requires balancing innovation with safety, strengthening sovereign digital capabilities, and ensuring inclusive global governance so that **AI remains a driver of sustainable and equitable development rather than a source of strategic inequality**.

Drishti Mains Question:

Agentic AI presents unprecedented opportunities as well as cybersecurity and governance risks. Critically examine.

Frequently Asked Questions (FAQs)

1. What is the UN's Independent International Scientific Panel on Artificial Intelligence?

It is a UN-backed expert body established to provide evidence-based scientific assessments on AI risks, opportunities, and governance to support informed global policymaking.

2. What is meant by the "AI Divide"?

The AI Divide refers to disparities in **compute infrastructure, AI talent, governance capacity, and application capability**, rather than merely unequal access to digital technologies.

3. What is Agentic AI, and why is it a governance concern?

Agentic AI consists of autonomous AI systems capable of planning and executing tasks with minimal human intervention, raising concerns over cybersecurity, accountability, and loss of human control.

4. Which Indian initiatives support responsible AI development?

Key initiatives include the **IndiaAI Mission**, **BHASHINI**, **BharatGen**, **Ayushman Bharat Digital Mission (ABDM)**, **CERT-In**, **I4C**, and the **Digital Personal Data Protection (DPDP) Act, 2023**.

5. Why is multilingual AI important for India?

Multilingual AI enables inclusive digital access, improves AI accuracy across Indian languages, reduces algorithmic bias, and supports initiatives such as **BHASHINI** and **Digital Public Infrastructure (DPI)**.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/8mSI0G_2uWg](https://www.youtube.com/embed/8mSI0G_2uWg)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. With the present state of development, Artificial Intelligence can effectively do which of the following? (2020)

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

- (a) 1, 2, 3 and 5 only
- (b) 1, 3 and 4 only
- (c) 2, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

Ans: (b)

Mains

Q. Discuss different types of cyber crimes and measures required to be taken to fight the menace. (2020)

India Seeks Clarity on 'Climate Tipping Points' at Bonn Climate Conference 2026



Source:TH

At the **64 th Sessions of the Subsidiary Bodies (SB64) to the UNFCCC** , held in Bonn, Germany, India urged scientific caution against incorporating the term "climate tipping points" into international climate negotiations without clear definitions and robust scientific consensus.

- The mid-year negotiations aimed to lay the technical groundwork for the upcoming **COP31 UN Climate Change Conference** in Antalya, Türkiye.
- **Climate Tipping Points:** A tipping point is a **critical threshold** in the Earth's climate system which, when crossed, leads to **rapid, self-reinforcing, and potentially irreversible changes** , independent of the linear pace of greenhouse gas emissions.
- **Major Global Examples:** Identified tipping points include the **dieback of the Amazon rainforest** into a dry savanna, the collapse of the **Atlantic Meridional Overturning Circulation (AMOC)** , widespread **coral reef bleaching** , and drastic shifts in the **Indian and West African monsoon systems** .
- **India's Stand on Scientific Rigor:** India argued that the concept lacks a universally accepted scientific definition.
 - The delegation highlighted that significant uncertainties remain regarding the **exact thresholds, the timing of occurrence, and the magnitude of impacts** , cautioning that climate policy must remain grounded in robust, peer-reviewed scientific evidence rather than ambiguous terminology.
- **The 1.5°C Misconception:** The talks brought to light a prevalent misunderstanding—the **1.5°C and 2°C limits set under the Paris Agreement are political temperature targets** , and not definitive, scientifically proven climate tipping points.
- **The EU's Pushback:** The **European Union** opposed India's stance, characterizing the call for clarity as "coordinated misinformation" and an "obstruction" that could delay urgent climate action and create confusion during policy discussions.
- **Implications for Developing Nations:** For the Global South, adopting poorly defined thresholds into **UNFCCC mandates** risks forcing disproportionate mitigation burdens onto developing economies, potentially undermining the core principles of **Equity and Common But Differentiated Responsibilities (CBDR)** .
- **Climate Finance & Trade Measures:** In addition to the tipping points debate, India raised alarms over **declining climate finance from developed countries** and pushed back against unilateral trade policies like the EU's **Carbon Border Adjustment Mechanism (CBAM)** , arguing they unfairly penalize developing nations transitioning away from poverty.

Bonn Climate Change Conference

- The **Bonn Climate Change Conference** is the **mid-year climate summit** held under the **UNFCCC** . Formally known as the **Sessions of the Subsidiary Bodies (SBs)** , it is one of the two regular UNFCCC meetings alongside the annual **Conference of the Parties (COP)** .
 - Held annually ahead of the **COP** , it reviews the implementation of climate commitments, advances technical negotiations, and prepares the agenda for the next COP.
- It is supported by two bodies: the **Subsidiary Body for Implementation (SBI)** , which oversees implementation and support, and the **Subsidiary Body for Scientific and Technological Advice (SBSTA)** , which provides scientific guidance based on **Intergovernmental Panel on Climate Change (IPCC)** assessments.

Read more: [Bonn Climate Change Conference 2025](#)

eSARAS: Empowering Rural Women through Digital Markets



Source: PIB

eSARAS is a key initiative driving women's entrepreneurship, preserving India's cultural heritage, and building a digitally empowered, self-reliant rural economy.

- **About:** eSARAS, developed by the **Ministry of Rural Development under DAY-NRLM** , is an official online marketplace created exclusively for products made by women **Self-Help Groups** and their federations.
- **Purpose:** The platform supports rural artisans such as **weavers, potters, marble artisans, Pashmina producers, and women-led cooperatives** by giving them direct access to buyers across India.
- **Product Range:** It lists products such as home and living items, apparel and accessories, personal care products, food items, kids' toys and crafts like Chanderi sarees, marble crafts and Pashmina products.
- **Institutional Support:** eSARAS supports marketing, branding, logistics, processing, packaging and shipping of women-led rural enterprise products.
- **Platform Evolution:** eSARAS has grown from an e-commerce portal into a wider ecosystem with a mobile app, fulfilment centre, ONDC integration, **UMANG listing, SARAS Shakti Collection** and SARAS Aajeevika Gallery in New Delhi.
- **Significance:** eSARAS strengthens women-led rural entrepreneurship, digital inclusion, market access, cultural heritage preservation, sustainable livelihoods and inclusive rural economic growth.

DAY-NRLM

- Deendayal Antyodaya Yojana - National Rural Livelihood Mission (DAY-NRLM) is a flagship poverty alleviation program being implemented by the **Ministry of Rural Development** to reduce poverty by enabling poor households to access gainful self-employment and skilled wage employment opportunities **resulting in sustainable and diversified livelihood options for the poor.**
- DAY-NRLM aims at mobilizing rural poor households into Self Help Groups (SHGs) and continuously nurturing and supporting them to take **economic activities till they attain an appreciable increase in income over a period** of time to come out of abject poverty.



Read more: [Women-Led Economic Growth for Viksit Bharat 2047](#)

Exercise RIMPAC



Source: TH

The **Indian Navy's P-8i maritime patrol aircraft** is participating in the **Rim of the Pacific Exercise(RIMPAC) 2026** in Hawaii, reflecting India's growing role in Indo-Pacific maritime cooperation.

- **About RIMPAC:** The RIMPAC is the **world's largest international naval exercise**, hosted by the U.S. Pacific Fleet, and has been conducted since 1971.
 - The exercise strengthens **interoperability, Maritime Domain Awareness, readiness, and coordination** among participating naval forces.
- **Theme:** RIMPAC 2026 is the 30th edition of the exercise, held under the theme **"Partners: Integrated and Prepared."**
- **Participation:** The exercise will involve 30 nations, more than **30 surface ships, five submarines and over 200 aircraft, with major participants including India, the United States, Australia, Japan, France, the United Kingdom, South Korea, Singapore and Indonesia.**
- **Training Areas:** RIMPAC includes **complex maritime drills such as anti-submarine warfare, air defence, amphibious operations, gunnery and missile exercises**, counter-piracy, mine countermeasures, and diving and salvage operations.
- **Significance:** By participating in RIMPAC, the Indian Navy enhances its **ability to operate with allied and partner navies** while contributing to sea-lane security and maritime stability in the Indo-Pacific.

P-8i Maritime Patrol Aircraft

- The P-8I is a **multi-role, long-range maritime reconnaissance** and anti-submarine warfare aircraft based on the Boeing 737-800 platform.
- It supports **maritime surveillance, strike missions, electronic warfare**, search and rescue, targeting support for weapon systems, and anti-submarine operations. It was also used for reconnaissance and surveillance during the post-Galwan high-altitude stand-off.
- The **aircraft can carry air-to-ship missiles** and torpedoes. The Indian Navy currently operates 12 P-8I aircraft and plans to acquire six more.

Read more: [India's Maritime Strategy](#)